

Supplementary Materials

Auditing Protein–Protein Interaction Benchmarks by Relation Pattern and Entity Boundary

These materials support the main paper. The tables below give the full evidence-tier, matched-carrier, residual-decomposition, and within-corpus results summarized in the main text.

S1 Evidence tiers and audit scope

Table S1 assigns each released positive to its highest evidence tier. Committed coverage is $1,683/4,187 = 40\%$; structural or connected-path evidence covers $4,131/4,187 = 99\%$. These positive-only counts do not define the statistical population: all 4,184 candidate-relation observations are analyzed, including non-positives and review-only statements.

The “held-as-evidence” tier ($302 = 7\%$) is deliberate abstention: the reader saw an event or relation but did not commit it as a pair. Generic is-a, taxonomy and membership statements stay in this tier. Role-nominal is-a forms such as A is an activator of B are committed only when the role noun is an interaction-like trigger and the endpoint is introduced by of or for. Of the held tier, 130 are appositive is-a relation statements. The path tier ($2,146 = 51\%$) requires a nonempty saved dependency path. It is a retrospective diagnostic queried at known positive endpoints, not label-free discovery or a confidence score. Empty-path diagnostics, including search-limit failures, do not count: 56 positives have neither structural nor connected-path evidence. Their identifiers are included in the repository.

Table S1: Coverage by evidence type. Columns are mutually exclusive highest-tier assignments per released positive. The evidence columns sum to 4,131 of 4,187 positives. Counts without structural or connected-path evidence are 3, 47, 0, 5 and 1 for AIMed, BioInfer, HPRD50, IEPA and LLL, respectively.

corpus	rel. pos.	clause/SVO	event-nom.	complex/state	held-as-ev.	path-only ev.
AIMed	991	416 (42%)	78 (8%)	47 (5%)	39 (4%)	408 (41%)
BioInfer	2534	612 (24%)	132 (5%)	80 (3%)	219 (9%)	1444 (57%)
HPRD50	163	91 (56%)	12 (7%)	0 (0%)	7 (4%)	53 (33%)
IEPA	335	117 (35%)	25 (7%)	0 (0%)	27 (8%)	161 (48%)
LLL	164	67 (41%)	6 (4%)	0 (0%)	10 (6%)	80 (49%)
total	4187	1303 (31%)	253 (6%)	127 (3%)	302 (7%)	2146 (51%)

The retained parser ablation disables QANom and coreference together. Of the originally significant comparisons across the six separately corrected families, 61 remain significant; the ablated run has 51 isolated-contrary cells. These are sensitivity counts across families, not replacements for the primary-family result or an isolated estimate of QANom’s effect. The corresponding retained outputs are listed in the reproduction manifest.

S2 Sensitivity to the inclusion rule

Table S1 grades released positives by evidence. It cannot be used to subset the statistical population: keeping only strongly-evidenced positives while keeping every negative would move each corpus’s positive rate by construction, and that rate is the quantity under test. One inclusion rule can be varied without that circularity, namely the reader’s own commitment. It is recorded per observation for positives and negatives alike, and it states whether an observation was applied as a pair decision, held for review, not applied, or not selected. The main analyses do not filter on commitment, so they use the widest of these rules. Table S2 recomputes them under three narrower rules.

The effect sizes are stable and the significance counts are not. The mean absolute positive-rate difference across tested comparisons stays between 0.29 and 0.32, and the signed AIMed-to-BioInfer contrast stays between 0.235 and 0.272. Meanwhile the number of corrected-significant comparisons falls from 14 of 54 to 7 of 41, as the population shrinks by a third. Fisher tests on smaller cells lose power. The effect-

size summaries remain similar, but the tested comparison sets change with support; this is a sensitivity analysis, not proof of invariance. Under every inclusion rule, the within-corpus AUC-lift interval includes zero. No rule can address the observations the reader never emits at all, which the main text states as a limitation.

Table S2: Headline results under four inclusion rules, widest first. Rules are defined on the reader’s per-observation commitment, not on the positives-only evidence tiers of Table S1. The AIMed–BioInfer column gives the signed contrast with the number of matched constructions in parentheses. The first row is the population every result in the paper uses and reproduces its numbers. “mean $|\Delta|$ ” is the mean absolute difference in released-positive rate across the comparisons evaluable under each rule; the set can change with support, and the within-corpus lift carries its document-cluster bootstrap interval.

Inclusion rule	obs.	primary family	mean $ \Delta $	within-corpus lift	AIMed–BioInfer
Every observation (reported)	4,184	14/54	0.324	+0.002 [−0.005, 0.010]	0.267 (30)
Drop not-selected	4,134	13/53	0.312	−0.001 [−0.009, 0.008]	0.272 (30)
Applied or review-only	3,625	10/48	0.293	−0.003 [−0.011, 0.005]	0.235 (28)
Applied only	2,781	7/41	0.301	+0.001 [−0.012, 0.014]	0.239 (25)

S3 Matched-carrier robustness

The main paper reports that the cross-corpus positive-rate gap persists after matching the recorded carrier. Table S3 gives the full ladder in both forms.

The unpaired columns condition on progressively finer carrier information and recompute the gap on whatever strata still have shared support. That set shrinks from 35 to 20, so those four means are computed on four different populations and their differences cannot be read as an effect of carrier conditioning. The sequence is also not monotone. The paired column fixes this. It restricts every level to the 18 base constructions that survive all four levels, and averages a construction’s carrier sub-strata before comparing, so each level contributes one value per construction over identical units. On that footing the gap does not fall at all ($0.324 \rightarrow 0.339 \rightarrow 0.357 \rightarrow 0.357$). Each of those three changes, measured against the unconditioned level, has a bootstrap interval that straddles zero.

Matching uses the untruncated predicate, construction type, polarity, argument shape, and case marker. The unconditioned gap is 0.364 over 35 strata. For each stratum, the gap is the maximum minus minimum released-positive rate among eligible corpora; these ranges are averaged across strata. This multi-corpus range differs from the signed BioInfer-minus-AIMed contrast reported in Table S4. Carrier-class codebook coverage is $1,471/1,972 = 74.6\%$ of carrier-bearing rows.

Table S3: Matched cross-corpus gap as carrier conditioning is added, unpaired (on all strata that remain evaluable) and paired (on the 18 base constructions common to all four levels, with bootstrap intervals over constructions).

matching condition	strata	unpaired gap	paired gap (95% CI)
construction only (no carrier)	35	0.364	0.324 (0.21–0.45)
+ carrier class	28	0.362	0.339 (0.23–0.47)
+ carrier word	22	0.395	0.357 (0.26–0.48)
+ carrier word $\times$ role	20	0.352	0.357 (0.26–0.48)

S4 Nulls for the matched gap

Two nulls address different explanations for the matched gap (Table S4). A label permutation within each construction preserves that construction’s positive total and each corpus’s support, and destroys any per-corpus difference. Its mean gap of 0.158 is the small-sample baseline that the range statistic carries even with no corpus effect at all. This null is descriptive only, because it also destroys the corpus base rates and does not isolate construction-specific variation beyond prevalence.

The inferential null is the additive offset. Labels are simulated from $\text{logit}(p) = \alpha_{\text{corpus}} + \beta_{\text{construction}}$, with no corpus-by-construction interaction. Rejecting it therefore means that corpora do not merely differ by one global level of strictness. The null is fitted unpenalized. Documents are resampled within corpus and the model is refitted on each draw, with a document random intercept calibrated to the observed residual intra-document correlation ($\text{ICC} = 0.087$, mean cluster about six rows). Without that random intercept the null would draw rows independently, even though a single BioInfer sentence can contribute up to 36 observations, and its p -values would be anticonservative. The offsets are estimated on data that still contains the interaction, so calibration remains conditional on the model assumptions rather than exact.

For inference, we compute the sample standard deviation of the signed BioInfer-minus-AIMed rate differences across constructions. This compares one fixed corpus pair throughout and measures variation beyond its mean contrast. The descriptive mean and the tested standard deviation (“spread”) are reported separately.

Table S4: Signed BioInfer-minus-AIMed contrast over matched constructions, against the document-clustered additive-offset null. The mean contrast is what a global offset already predicts; the evidence for construction-specific treatment is the spread.

matching condition	strata	mean	spread	null spread (p)
construction only	30	+0.267	0.277	0.199 (0.009)
+ carrier class	23	+0.263	0.328	0.187 (<0.001)
+ carrier word	19	+0.295	0.345	0.189 (<0.001)
+ carrier word $\times$ role	17	+0.232	0.300	0.182 (0.003)

Table S5 gives descriptive examples grouped more coarsely by trigger, verbal or nominal form, and asserted polarity. These are not the full matching strata used in Table S4 or the primary canonical-construction tests.

Table S5: Descriptive trigger groups with asserted polarity. Rates are followed by [positives/support]; verbal and nominal denote forms, not grammatical voice. These groups do not control argument shape or case marker.

Trigger and form	Positive rate by corpus
associate (verbal)	AIMed 0.82 [28/34], BioInfer 0.67 [6/9], HPRD50 1.00 [5/5]
inhibit (verbal)	AIMed 0.20 [3/15], BioInfer 0.57 [12/21], LLL 1.00 [7/7]
activate (verbal)	AIMed 0.62 [16/26], BioInfer 0.75 [6/8], IEPA 1.00 [5/5], LLL 1.00 [5/5]
link (verbal)	AIMed 0.00 [0/5], BioInfer 0.96 [23/24]
associate (nominal)	AIMed 0.45 [17/38], BioInfer 0.89 [16/18]

S5 Residual corpus effect: full decomposition

Held-out 5-fold AUC, encoders refit in-fold, document-clustered where noted ($n = 4,184$ pattern observations, positive rate 0.500).

Model specification

Categorical cues are one-hot encoded with the encoder fitted inside each training fold, so a test fold’s category inventory never leaks; categories occurring fewer than ten times in a fold are pooled. Predicate cardinality is capped at the twenty-five most frequent lemmas, for the predictive models only. That screen uses the full analysis population without labels and runs before fold assignment. Unlike the encoder and the bag-of-words vocabulary, it is therefore not a fully held-out preprocessing step, and we state this rather than imply otherwise. The matched-construction analyses use the untruncated predicate instead, because pooling infrequent triggers under one symbol would build strata that are not in fact matched on the trigger.

Models are L_2 -regularized logistic regressions ($C = 1$). The vocabulary control is a binary bag of words over the sentence, with protein markers removed. It is fitted inside each fold, with a minimum document frequency of ten and a cap of 800 features. Table S6 reports 400 and 1,500 features as a sensitivity check. Folds are stratified five-fold with a fixed seed, or grouped by document where noted.

Table S6: Residual corpus effect and its controls. Every figure is a mean over five folds with its standard error, except the within-corpus lift, which uses document-grouped folds with a document-cluster bootstrap interval in brackets; lifts are paired, in the sense that both models are scored on identical folds.

model / control	held-out AUC
corpus-only baseline	0.664 ± 0.003
semantic cues only (m_1)	0.738 ± 0.005
semantic cues + corpus (m_2)	0.804 ± 0.002
corpus lift over cues	$+0.066 \pm 0.005$
Prevalence controls	
cues + corpus base rate only (m_3)	0.803 ± 0.002
lift over cues	$+0.065 \pm 0.005$
lift, within each corpus (document-grouped)	$+0.002 [-0.005, 0.010]$
Vocabulary control	
cues + sentence content (bag-of-words)	0.817 ± 0.003
cues + content + corpus	0.839 ± 0.005
corpus lift over cues + content	$+0.022 \pm 0.003$
same, document-clustered folds	$+0.066 \pm 0.008$
content vocabulary sweep (400 / 800 / 1500 features)	$+0.032 / +0.022 / +0.022$

Two readings follow, and the first is the more important. The corpus indicator is close to a per-corpus intercept, because released-positive rates in this population run from 0.33 (AIMed) to 0.91 (LLL). Replacing the indicator with that single number reproduces the lift almost exactly ($+0.065$ against $+0.066$). Recomputing AUC inside each corpus, where an intercept cannot change any ranking, needs document-grouped folds rather than a plain pair-level split, because one BioInfer sentence holds up to 36 candidate pairs. On document-grouped folds, with a document-cluster bootstrap in place of the five-fold standard error, the within-corpus lift is $+0.002 [-0.005, 0.010]$, which does not exclude zero. This bootstrap resamples fixed out-of-fold predictions; models are not refitted per replicate, so the interval is conditional on the fitted folds. The aggregate corpus effect is therefore mostly base-rate memorization, and the paper’s claim rests on the construction-level analyses instead.

Second, the bag-of-words control absorbs most of the raw lift only when folds are unclustered ($+0.066 \rightarrow +0.022$). With folds grouped by document it absorbs almost nothing ($+0.071 \rightarrow +0.066$). The cue-only lift is essentially unchanged by clustering, so this is compatible with removal of document-level vocabulary leakage, but does not establish that mechanism or exclude changes in estimation uncertainty. Under unclustered folds, observations from the same document can enter both training and test sets. Document-specific vocabulary could then absorb corpus signal; these results do not separate that possibility from other effects of fold construction. Document-grouped evaluation is therefore the more informative control for transfer to unseen documents.

Table S7 gives the complete crossed design, so that the effect of restricting to two corpora can be separated from the effect of clustering.

Table S7: Corpus lift (AUC gain over the same cue-only model) with sample and fold construction crossed against the vocabulary control.

sample and folds	cues only	cues + bag of words
full five-corpus sample	$+0.066 \pm 0.005$	$+0.022 \pm 0.003$
document-clustered folds	$+0.071 \pm 0.008$	$+0.066 \pm 0.008$
two largest corpora (AIMed + BioInfer)	$+0.062 \pm 0.003$	$+0.020 \pm 0.002$
document-clustered folds	$+0.068 \pm 0.008$	$+0.058 \pm 0.002$

S6 Within-corpus inconsistency

The within-corpus check asks whether a single corpus labels the same canonical construction uniformly while holding source-corpus membership fixed. Vocabulary, semantics, and dependence can still vary within a corpus. Table S8 gives the per-corpus breakdown on the canonical pattern dimension. Of the 169

assessable cells, 62 are unanimous, 53 split, and 54 are isolated-contrary (both labels occur and the minority share is at most 0.25). The null permutes released labels within each corpus, preserving that corpus’s positive total and every construction’s firing count, and recounts isolated-contrary cells; corpora are never pooled. Over 10,000 permutations (seed 0, support ≥ 5 , minority share ≤ 0.25) the expected count is 42.1 against the observed 54 ($p = 0.014$). AIMed and BioInfer, with the most recurring constructions, carry 27 and 22 of the 54. This permutation does not preserve document-level label dependence and should be read as a descriptive check.

Table S8: Within-corpus inconsistency, canonical pattern dimension.

corpus	evaluable	mixed share	assessable	unanimous	isolated-contrary	split
AIMed	1490	0.89	74	22	27	25
BioInfer	1365	0.84	76	27	22	27
HPRD50	83	0.13	8	6	2	0
IEPA	60	0.60	7	3	3	1
LLL	24	0.00	4	4	0	0
total	3022	n/a	169	62	54	53

S7 Reproduction artifacts

This submission package contains the main paper PDF, this supplementary PDF, and a consolidated pattern/carrier review workbook: every pattern, carrier, carrier $\times$ pattern, and trigger cell with support, counts, and positive rate; the cross-corpus comparisons with BH significance; and the within-corpus summary. `annotation_items.csv` and `annotator_ratings.csv` link the fixed validation sentences to the anonymous four-question human ratings. Free-text annotation comments are excluded; the accompanying data dictionary specifies the answer categories and scoring denominators. The code and this appendix are available at <https://github.com/lymn3010/ppi-benchmark-audit>. The repository includes the reproduction code, the pair-level derived ledgers, and the blinded validation instrument.

S8 Human validation and instrument development

The fixed sample contains 163 observations: 46 AIMed, 44 BioInfer, 32 HPRD50, 22 IEPA, and 19 LLL. Sampling was stratified by corpus, carrier class, and membership in a corrected-significant comparison (65 observations versus 98 controls). One independent annotator rated all items, seeing the sentence, marked proteins, and reader trigger and carriers while blinded to released labels, cell rates, and significance. The questions assessed relation assertion, trigger correctness, carrier attachment, and whether the marked proteins were the relation’s endpoints. Estimates are unweighted and restricted to definite responses, not population accuracy estimates.

The annotator found that carrier examples illustrated parts and attributes too narrowly. Clarified instructions also covered processes, roles, and reagents; reassessment yielded carrier agreement of 0.61 (95% CI 0.53–0.68). The endpoint question had 129 definite responses, which is not the number of affirmative relation judgments. It lacked a not-applicable option, limiting interpretation when no relation was asserted. Endpoint responses therefore must not be reported as reader precision.

The significant-cell sample covers nine constructions, mostly with fewer than ten observations each, and cannot validate individual cells. Non-significant corpus differences are not equivalence tests, and no inter-rater agreement is claimed. The fixed sample, instrument, anonymous ratings, and scoring code are included in the reproduction materials (`validation/README.txt` and its associated files under `supplements/` in the repository).